

NIDUBROLU PAVAN SANDEEP

Frontend developer, UI/UX, GenAI, Prompt Engineer

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PROFESSIONAL SUMMARY

Front-End Developer with hands-on experience in React.js, TypeScript, JavaScript, Redux Toolkit, React Router, Vite, HTML, CSS, and Tailwind CSS. Skilled in building responsive web applications, reusable component architectures, API integration, state management, authentication systems, and enterprise-grade user interfaces. Experienced in collaborating with cross-functional teams to develop scalable solutions and improve application performance. Passionate about modern frontend development, user experience, and writing clean, maintainable code.

Gen AI and Prompt Engineering professional with experience in prompt design, dataset management, and Natural Language Processing (NLP). Passionate about developing AI-driven solutions, optimizing model performance, and contributing to innovative projects.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python

Frontend Technologies: React.js, React Router, Next.js, Redux, Tailwind CSS, HTML5, CSS3, React Flow.

AI/ML Technologies: Gen AI, Prompt Engineering, LangChain, LLM Models, OpenCV, ML, YOLO

Database Systems: MySQL, Supabase

Development Tools: VS Code, Cursor, Postman, Figma, ChatGPT, Claude AI, DeepSeek

Version Control & Deployment: Git, Bitbucket, GitHub Actions, SonarQube, JIRA

Operating Systems: Linux, Windows

Other Skills: Code Review, Software Testing, Progressive Web Apps, Performance Optimization

WORK EXPERIENCE

BitCot Technologies Pvt Ltd, Chennai

(Jan 2025 – Jan 2026)

Front-End Developer / UI/UX

Project 1: Medical Image Transfer [LSR – Light Source Research]

Role: Front-End Developer

Development of a web-based medical imaging platform for analyzing and visualizing spine related diagnostic data using DICOM medical images.

- Developed a Next.js-based medical imaging platform for viewing and analyzing spine-related DICOM images.
- Integrated and customized Cornerstone.js to render and interact with medical images directly in the browser.
- Implemented imaging tools such as zoom, pan, measurements, annotations, and window-level adjustments for X-Ray, CT, and MRI analysis.
- Utilized Redux for centralized state management and efficient handling of viewer and image data.
- Integrated a 3D DICOM Viewer to enhance visualization of anatomical structures and diagnostic workflows.
- Built reusable React components and optimized application performance for large medical imaging datasets.

Technologies: Next.js, React, Redux, JavaScript, Cornerstone.js, DICOM, 3D DICOM Viewer, REST APIs

Project 2: PegDesk [Shaping the Future]

Role: Front-End Developer

Developed and enhanced a React.js-based platform for managing commercial cleaning and facility maintenance operations.

- Worked on a class-based React architecture, contributing to feature development, bug fixes, and maintenance of legacy code.
- Built and improved user interfaces for service request management, workforce scheduling, and client communication workflows.
- Developed reusable UI components and optimized application performance to improve user experience.
- Collaborated with cross-functional teams to implement business requirements and deliver platform enhancements.

Technologies: React.js, JavaScript, HTML, CSS, REST APIs, Redux

Project 3: TrackPro [Fleet Management & GPS Tracking System]

Role: Front-End Developer & UI/UX Design

Developed a real-time fleet management and GPS tracking platform for monitoring vehicle locations, status, and operational data.

- Built real-time dashboards to monitor vehicle location, status, speed, mileage, fuel levels, and trip data.
- Developed modules for vehicle and driver information, tracking operational metrics and fleet activity.
- Utilized React.js, TypeScript, and Redux to build scalable and responsive user interfaces.
- Created reusable components and optimized state management across tracking and reporting features.
- Enhanced user experience with GSAP and Framer Motion animations and responsive Bootstrap based layouts.

Technologies: React.js, TypeScript, Redux, GSAP, Framer Motion, Bootstrap, JavaScript, REST APIs, Reusable Component Architecture

Photon Interactive Pvt Ltd, Chennai

(Mar 2024 – Dec 2024)

Prompt Engineer / GenAI Developer (Internship / Apprentice)

Project : AI Interviewer

Intelligent automated interview assessment system with computer vision and AI analysis

- Designed and optimized prompts to improve the accuracy and relevance of AI-generated responses.
- Collaborated with developers to integrate AI-powered features into applications and internal tools.
- Implemented real-time eye tracking and gaze estimation techniques using OpenCV, Media Pipe, and YOLO.
- Contributed to improving model efficiency and user experience through testing, debugging, and iterative enhancements.
- Evaluated AI model outputs and refined prompts to enhance response quality and consistency.

Technologies: Python, OpenCV, MediaPipe, YOLOv5, Computer Vision, Prompt Engineering, Generative AI, NLP, Large Language Models (LLMs), Machine Learning, Deep Learning, Git, REST APIs

EDUCATION

Bachelor of Engineering – Computer Science Engineering

2020 – 2024

Sathyabama Institute of Science and Technology, Chennai

- CGPA: 8.73/10